LONDON METROPOLITAN UNIVERSITY

islington college
(इस्लिङ्टन कलेज)

Module Code & Module Title
CS6P05NI Project
Proposal

5% Individual Coursework
Submission: Final Submission
AY 2025 2026
Academic Semester: Autumn Semester 2025

Credit: 30 Credit Year Long Module

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Assignment Due Date: Wednesday, December 3, 2025
Assignment Submission Date: Wednesday, December 3, 2025
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I confirm that I understand my coursework needs to be submitted online via MST Classroom under the relevant module page before the deadline for my assignment to be accepted and marked. I am fully aware that late submissions will be treated as non-submission and a mark of zero will be awarded.

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1. Introduction

Agriculture plays a hypercritical role in fulfilling food security and economic stability globally. However, as the global awareness and need is growing for organic and healthy products the demand for efficient and sustainable farming has become more critical. Traditional farming still heavily depends on manual monitoring of the field and its environment which is efficient for very small-scale farming but is almost impossible in large farms which in return delays any problem detection reducing crop yield.

With rapid advancement in IoT (Internet of Things) and AI (Artificial Intelligence), old agriculture can be transformed using real-time data collection and analysis of the said data to get insight on the status of the field. This project adheres to this principal proposing a comprehensive smart agriculture system for monitoring critical field parameters to enhance farm productivity and sustainability.

1.1. Problem Statement:

"Modern agriculture is still heavily dependent on manual monitoring and labor-intensive crop management. Farmers struggle to manage soil nutrient (mostly NPK) balance as it is either too expensive or just unavailable for them leading to inefficient yield."

1.2. Project Statement:

"This project aims to create an intelligent IoT and AI powered **Smart Agriculture Field Monitoring and Management System**" and to so it certain goals need to be achieved that is:

- Monitor environment conditions
- Analyze data patterns using AI
- Provide real-time alerts via a dashboard.
- Provide suggestions for better yield of plants.

By integrating the capability of IoT to collect data and the analysis of AI the project visons to transform farms into precise agriculture system.

1.3. Project Features:

a. Multi Parameter Environment Monitoring:
   - Continuous monitoring of soil moisture levels at certain intervals to ensure plant hydration.
   - Tracking temperature and humidity.
   - Measure sunlight intensity.

b. Soil Analysis:
   - Measure soil PH levels to determine acidity or alkalinity for crops.
   - Monitor Nitrogen Phosphorous Potassium (NPK) levels for precise fertilization for the field.

c. AI-powered Analytics:
   - Use ML model to predict optimal irrigation schedule based on historical and real time data.
   - Analyze the NPK level for optimal fertilization rate for the field.

d. Cloud based data storage and Visualization:
   - Store all the data collected from sensors securely in cloud platforms.
   - Intuitive dashboard for real time data, historical patterns and insights for user.
   - Multiuser access for the platform.

e. Scalable and modular architecture:
   - Modular expansion support with additional sensors or actuators as per the need.
   - Flexible communication protocol for farm’s in all sort of areas.

1.4. Future Features:

- Image Based Crop Health Analysis
- Livestock monitoring
- Mobile application with offline capabilities.
- Integration with agriculture marketplace

2. Aims and Objectives

Aim: The aim of the project is to implement an IoT and AI enabled smart agriculture monitoring and management system that helps farmers with real time field monitoring, data analysis and insights to help maximize the resource optimization for the best possible growth of the crops.

Objectives:

- Deploy a comprehensive network of IoT sensors to continuously monitor critical parameters including soil moisture, temperature, humidity and soil PH and NPK value.
- Implement a microcontroller-based data acquisition system to safely transmit data into the cloud.
- Develop AI powered predictive analytics for identifying early crop stress level as well as give insights based on the data.
- Design a user-friendly UI/UX for to display real time reading and actionable insights for farmers.

3. Expected Outcomes and Deliverables

Expected Outcomes:

- An operational smart agriculture system capable of accurately measuring multiple environment parameters.
- Significant reduction in use of fertilizers through precise fertilizer calculation and usage.
- Growth in plants yields achieved through optimal growing conditions.
- Reduced manual labor by automating field monitoring which helps farmers.
- Early problem detection enabling identification of potential issues 2-5 days earlier than traditional methods.

Deliverables:

1. Complete hardware setup
2. Software and AI model.
3. User interface.
4. Comprehensive system documentation.

4. Project risks, threats and contingency plans

Table 1 Project risks, threats and contingency plans

Risk ID	Risk Description	Impact (1-5)	Likelihood (1-5)	Risk Level	Mitigation Strategy	Responsible Team	Contingency Plan
1	Sensor Failure or Inaccuracy	4	3	12	Use high-quality, calibrated sensors; implement regular calibration schedules; deploy redundant sensors for critical parameters.	Hardware Engineering Team	Maintain spare sensor inventory for immediate replacement; implement software algorithms to detect and flag sensor anomalies.
2	Internet Connectivity Issues in Remote Farms	4	4	16	Design system with local data storage capabilities; use LoRa or GSM/4G as	Network Engineering Team	Store data locally on microcontroller SD cards; sync automatically

					backup communication; implement offline mode with data synchronization.		when connectivity restores; use SMS-based alerts as fallback.
3	Power Supply Interruptions	4	3	12	Use solar panels with battery backup systems; implement low-power modes for microcontrollers; design power-efficient circuits.	Hardware Engineering Team	Deploy uninterruptible power supply (UPS) systems; implement automatic system restart protocols after power restoration.
4	Harsh Weather Damage to Hardware	5	3	15	Use IP67/IP68 rated weatherproof enclosures; elevate sensors above flood levels; implement lightning protection circuits.	Hardware Engineering Team	Design modular hardware for easy component replacement; maintain emergency repair kits; establish rapid response

							maintenance protocols.
5	AI Model Inaccuracy or Overfitting	3	3	9	Train models on diverse datasets from multiple farms and seasons; implement cross-validation; regularly update models with new data.	AI Development Team	Provide manual override options for farmers; implement confidence scoring for AI recommendations; allow user feedback to improve models.
6	Data Security and Privacy Breaches	4	2	8	Implement encryption for data transmission and storage; use secure authentication; follow data protection regulations.	Security Team	Establish incident response protocols; implement regular security audits; maintain encrypted backups; notify users of breaches promptly.

7	Cloud Service Downtime	3	2	6	Use reliable cloud providers with SLA guarantees; implement local backup servers; design system to function with cached data.	Cloud Engineering Team	Switch to backup cloud provider; use local Raspberry Pi as temporary server; maintain critical functions during downtime.
8	Integration Challenges Between Components	3	3	9	Follow modular design principles; use standard communication protocols; conduct early integration testing.	Software Engineering Team	Allocate dedicated integration testing time; maintain detailed API documentation; use middleware for compatibility.
9	Budget Overruns	3	3	9	Create detailed budget planning; prioritize essential features; explore cost-effective	Project Management Team	Implement phased development; use open-source software; partner with agricultural

					component alternatives; seek grants or subsidies.		universities for equipment sharing.

Risk Level	Range	Indicator
Extreme Risk	16-25	
High Risk	10-15	
Medium Risk	5-9	
Low Risk	1-4	

5. Methodology

The methodology chosen for the project after looking into different development approaches (models) like Agile, Waterfall etc. is **Iterative Prototyping**. It was chosen because it allows the system to be developed in small functions which is almost essential for a sensor heavy hardware project like this.

The model provides flexibility for both hardware and software integration as well as sensor calibration and environment testing which is critical for the features in the project as we need the approach of “test → fix → test → improve” as it is needed not only to perfect the hardware design but also for the repeated tuning needed for the AI model.

1) Phase 1: Requirement Gathering and Increment Planning

   Identify system requirements and breakdown of the project based on features.

   Actions:

   - Identify main requirements for the user.
   - Gather system requirement based on users needed features.
   - Conduct assessment based on agriculture use cases (survey).
   - Develop a product backlog of activity and tasks for the project.
   - Divide features into clear increments and define goals and criteria for each increment.

2) Phase 2: Initial System Design

   Prepare a preliminary system architecture both hardware and software.

   Activity:

   - Design a overall architecture for hardware and software component.
   - Create visual diagrams such as Data Flow Diagrams, Hardware wiring layout etc.
   - Plan for the integration between the increments.
   - Identify potential risks (mostly for the sensors and data from it).

3) Phase 3: Incremental Development and Prototyping

   Each increment follows the cycle:

Figure 1 Design Process

Activity:

- Increment 1: Basic Sensor Integration
- Increment 2: Sensor calibration
- Increment 3:data processing
- Increment 4:Real time dashboard interface
- Increment 5: Alerts and Recommendation system
- Increment 6L Final Integrated prototype.

4) Phase 4: Testing

   Testing is performed at every increment and also at the final integration.

   Actions:

   - Unit Testing
   - Integration Testing
   - Field Testing
   - Calibration Testing
   - User Evaluation

5) Phase 5: Review and Refine

Each increment's result is reviewed and updated on next one based on its findings.

Actions:

- Review performance of each increment.
- Identify issues.
- Update the design and backlog for the next increment.
- Ensure each increments criteria is met before continuation of increment.

6) Phase 6: Final Evaluation and Deployment

After completion of all the increments the system is validated.

Actions:

- Conduct full system testing.
- Validate accuracy and usability of the features.
- Prepare the final prototype.
- Documentation

6. Resource Requirements

1.1. Hardware

- ESP32 Dev-Board
- Capacitive Soil Moisture Sensor
- DHT22 Humidity and Temperature Sensor
- BH1750 Sunlight sensor
- Analog Soil pH Sensor with Signal Conditioning Module
- Soil NPK Sensor (RS485/Modbus Type)
- RS485 to TTL Converter Module
- Lithium-Ion or LiPo Battery Pack (7.4V or 11.1V)
- MicroSD Card Module
- Sensor Mounting Rods / Soil Probes
- Breadboard / Prototype PCB
- Jumper Wires (Male–Male, Male–Female)
- USB Cable for ESP32 Programming

1.2. Software

- Arduino Ide
- ESP32 board drivers
- Python
- Flask API
- Sensor Libraries

1.3. Development Tools

- NumPy & Pandas
- Matplotlib
- REST API integration tools
- VS Code/ Arduino IDE
- Multimeter for hardware testing
- Soldering iron and wire.
- Jupyter notebook
- Draw.io
- Canva

7. Work Breakdown Structure

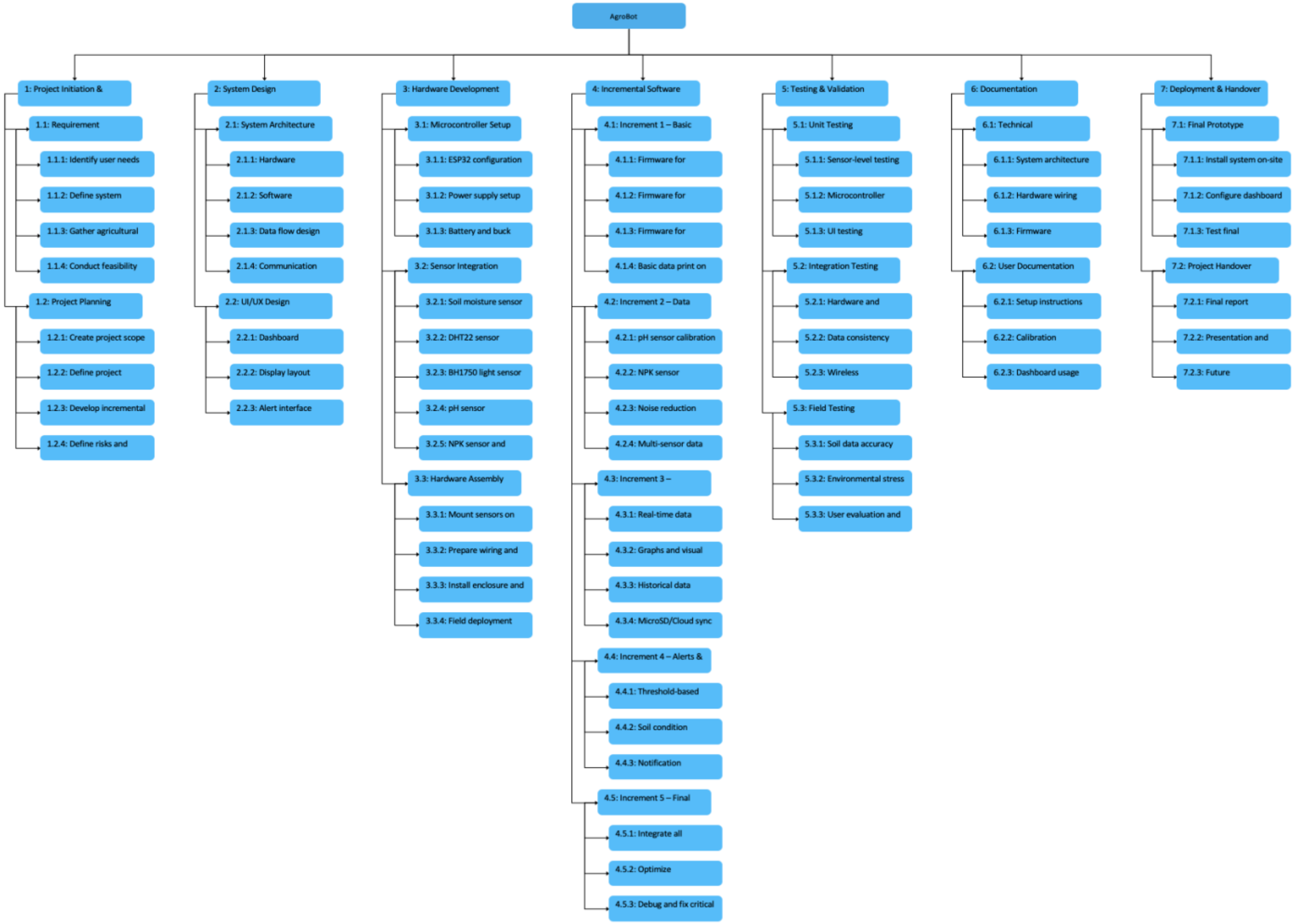
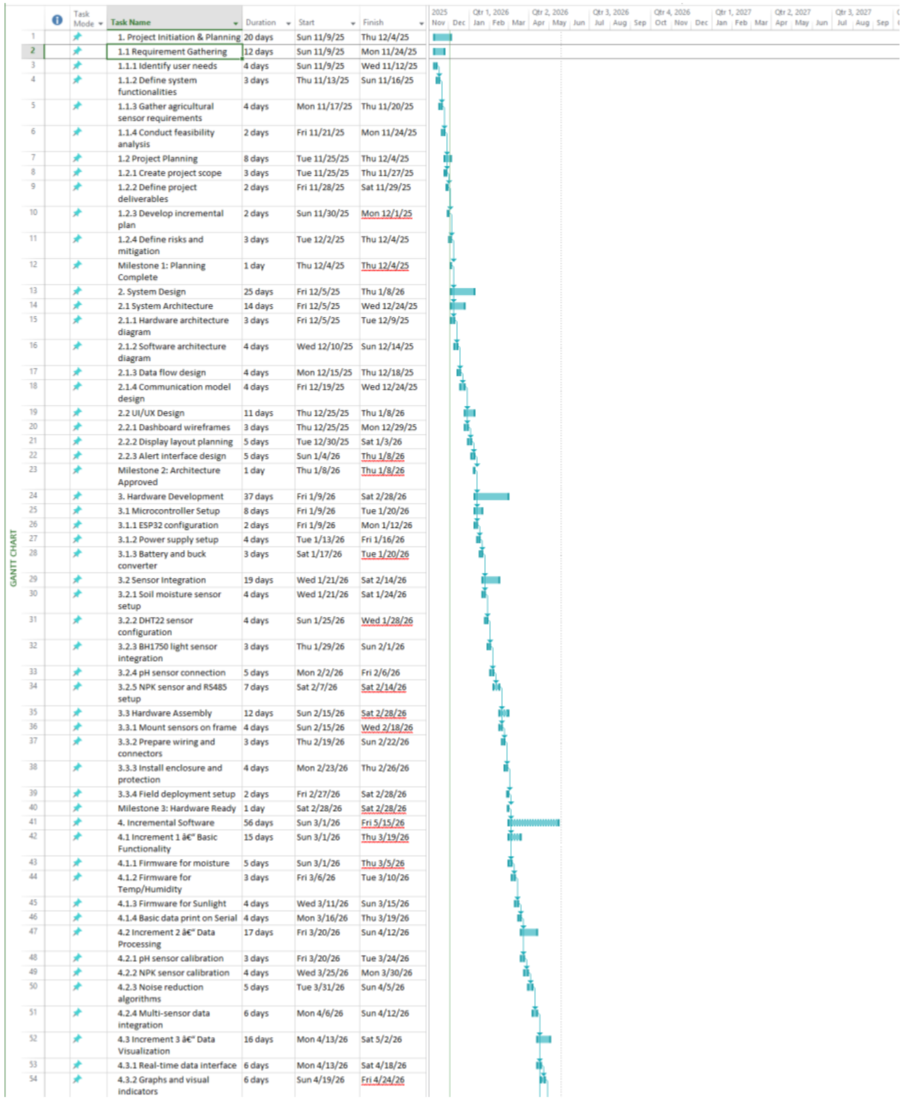


Figure 2 Work Breakdown Structure

8. Milestone

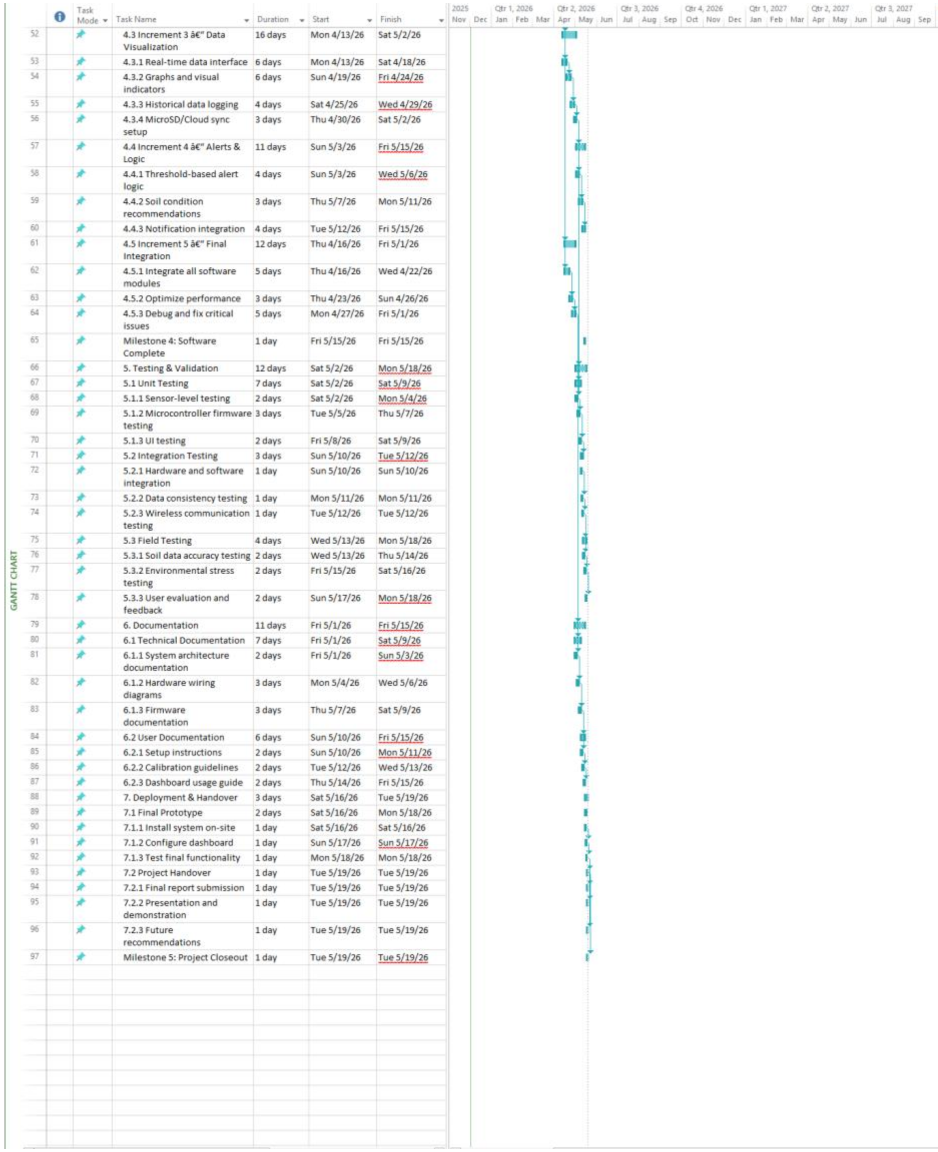
Figure 3 Milestones

9. Gantt Chart



ID	Task Mode	Task Name	Duration	Start	Finish
1		1. Project Initiation & Planning	20 days	Sun 11/9/25	Thu 12/4/25
2		1.1 Requirement Gathering	12 days	Sun 11/9/25	Mon 11/24/25
3		1.1.1 Identify user needs	4 days	Sun 11/9/25	Wed 11/12/25
4		1.1.2 Define system functionalities	3 days	Thu 11/13/25	Sun 11/16/25
5		1.1.3 Gather agricultural sensor requirements	4 days	Mon 11/17/25	Thu 11/20/25
6		1.1.4 Conduct feasibility analysis	2 days	Fri 11/21/25	Mon 11/24/25
7		1.2 Project Planning	8 days	Tue 11/25/25	Thu 12/4/25
8		1.2.1 Create project scope	3 days	Tue 11/25/25	Thu 11/27/25
9		1.2.2 Define project deliverables	2 days	Fri 11/28/25	Sat 11/29/25
10		1.2.3 Develop incremental plan	2 days	Sun 11/30/25	Mon 12/1/25
11		1.2.4 Define risks and mitigation	3 days	Tue 12/2/25	Thu 12/4/25
12		Milestone 1: Planning Complete	1 day	Thu 12/4/25	Thu 12/4/25
13		2. System Design	25 days	Fri 12/5/25	Thu 1/8/26
14		2.1 System Architecture	14 days	Fri 12/5/25	Wed 12/24/25
15		2.1.1 Hardware architecture diagram	3 days	Fri 12/5/25	Tue 12/9/25
16		2.1.2 Software architecture diagram	4 days	Wed 12/10/25	Sun 12/14/25
17		2.1.3 Data flow design	4 days	Mon 12/15/25	Thu 12/18/25
18		2.1.4 Communication model design	4 days	Fri 12/19/25	Wed 12/24/25
19		2.2 UI/UX Design	11 days	Thu 12/25/25	Thu 1/8/26
20		2.2.1 Dashboard wireframes	3 days	Thu 12/25/25	Mon 12/29/25
21		2.2.2 Display layout planning	5 days	Tue 12/30/25	Sat 1/3/26
22		2.2.3 Alert interface design	5 days	Sun 1/4/26	Thu 1/8/26
23		Milestone 2: Architecture Approved	1 day	Thu 1/8/26	Thu 1/8/26
24		3. Hardware Development	37 days	Fri 1/9/26	Sat 2/28/26
25		3.1 Microcontroller Setup	8 days	Fri 1/9/26	Tue 1/20/26
26		3.1.1 ESP32 configuration	2 days	Fri 1/9/26	Mon 1/12/26
27		3.1.2 Power supply setup	4 days	Tue 1/13/26	Fri 1/16/26
28		3.1.3 Battery and buck converter	3 days	Sat 1/17/26	Tue 1/20/26
29		3.2 Sensor Integration	19 days	Wed 1/21/26	Sat 2/14/26
30		3.2.1 Soil moisture sensor setup	4 days	Wed 1/21/26	Sat 1/24/26
31		3.2.2 DHT22 sensor configuration	4 days	Sun 1/25/26	Wed 1/28/26
32		3.2.3 BH1750 light sensor integration	3 days	Thu 1/29/26	Sun 2/1/26
33		3.2.4 pH sensor connection	5 days	Mon 2/2/26	Fri 2/6/26
34		3.2.5 NPK sensor and RS485 setup	7 days	Sat 2/7/26	Sat 2/14/26
35		3.3 Hardware Assembly	12 days	Sun 2/15/26	Sat 2/28/26
36		3.3.1 Mount sensors on frame	4 days	Sun 2/15/26	Wed 2/18/26
37		3.3.2 Prepare wiring and connectors	3 days	Thu 2/19/26	Sun 2/22/26
38		3.3.3 Install enclosure and protection	4 days	Mon 2/23/26	Thu 2/26/26
39		3.3.4 Field deployment setup	2 days	Fri 2/27/26	Sat 2/28/26
40		Milestone 3: Hardware Ready	1 day	Sat 2/28/26	Sat 2/28/26
41		4. Incremental Software	56 days	Sun 3/1/26	Fri 5/15/26
42		4.1 Increment 1 â€“ Basic Functionality	15 days	Sun 3/1/26	Thu 3/19/26
43		4.1.1 Firmware for moisture	5 days	Sun 3/1/26	Thu 3/5/26
44		4.1.2 Firmware for Temp/Humidity	3 days	Fri 3/6/26	Tue 3/10/26
45		4.1.3 Firmware for Sunlight	4 days	Wed 3/11/26	Sun 3/15/26
46		4.1.4 Basic data print on Serial	4 days	Mon 3/16/26	Thu 3/19/26
47		4.2 Increment 2 â€“ Data Processing	17 days	Fri 3/20/26	Sun 4/12/26
48		4.2.1 pH sensor calibration	3 days	Fri 3/20/26	Tue 3/24/26
49		4.2.2 NPK sensor calibration	4 days	Wed 3/25/26	Mon 3/30/26
50		4.2.3 Noise reduction algorithms	5 days	Tue 3/31/26	Sun 4/5/26
51		4.2.4 Multi-sensor data integration	6 days	Mon 4/6/26	Sun 4/12/26
52		4.3 Increment 3 â€“ Data Visualization	16 days	Mon 4/13/26	Sat 5/2/26
53		4.3.1 Real-time data interface	6 days	Mon 4/13/26	Sat 4/18/26
54		4.3.2 Graphs and visual indicators	6 days	Sun 4/19/26	Fri 4/24/26

Figure 4 Gantt Chart (Part-1)



	Task Mode	Task Name	Duration	Start	Finish
52		4.3 Increment 3 â€" Data Visualization	16 days	Mon 4/13/26	Sat 5/2/26
53		4.3.1 Real-time data interface	6 days	Mon 4/13/26	Sat 4/18/26
54		4.3.2 Graphs and visual indicators	6 days	Sun 4/19/26	Fri 4/24/26
55		4.3.3 Historical data logging	4 days	Sat 4/25/26	Wed 4/29/26
56		4.3.4 MicroSD/Cloud sync setup	3 days	Thu 4/30/26	Sat 5/2/26
57		4.4 Increment 4 â€" Alerts & Logic	11 days	Sun 5/3/26	Fri 5/15/26
58		4.4.1 Threshold-based alert logic	4 days	Sun 5/3/26	Wed 5/6/26
59		4.4.2 Soil condition recommendations	3 days	Thu 5/7/26	Mon 5/11/26
60		4.4.3 Notification integration	4 days	Tue 5/12/26	Fri 5/15/26
61		4.5 Increment 5 â€" Final Integration	12 days	Thu 4/16/26	Fri 5/1/26
62		4.5.1 Integrate all software modules	5 days	Thu 4/16/26	Wed 4/22/26
63		4.5.2 Optimize performance	3 days	Thu 4/23/26	Sun 4/26/26
64		4.5.3 Debug and fix critical issues	5 days	Mon 4/27/26	Fri 5/1/26
65		Milestone 4: Software Complete	1 day	Fri 5/15/26	Fri 5/15/26
66		5. Testing & Validation	12 days	Sat 5/2/26	Mon 5/18/26
67		5.1 Unit Testing	7 days	Sat 5/2/26	Sat 5/9/26
68		5.1.1 Sensor-level testing	2 days	Sat 5/2/26	Mon 5/4/26
69		5.1.2 Microcontroller firmware testing	3 days	Tue 5/5/26	Thu 5/7/26
70		5.1.3 UI testing	2 days	Fri 5/8/26	Sat 5/9/26
71		5.2 Integration Testing	3 days	Sun 5/10/26	Tue 5/12/26
72		5.2.1 Hardware and software integration	1 day	Sun 5/10/26	Sun 5/10/26
73		5.2.2 Data consistency testing	1 day	Mon 5/11/26	Mon 5/11/26
74		5.2.3 Wireless communication testing	1 day	Tue 5/12/26	Tue 5/12/26
75		5.3 Field Testing	4 days	Wed 5/13/26	Mon 5/18/26
76		5.3.1 Soil data accuracy testing	2 days	Wed 5/13/26	Thu 5/14/26
77		5.3.2 Environmental stress testing	2 days	Fri 5/15/26	Sat 5/16/26
78		5.3.3 User evaluation and feedback	2 days	Sun 5/17/26	Mon 5/18/26
79		6. Documentation	11 days	Fri 5/1/26	Fri 5/15/26
80		6.1 Technical Documentation	7 days	Fri 5/1/26	Sat 5/9/26
81		6.1.1 System architecture documentation	2 days	Fri 5/1/26	Sun 5/3/26
82		6.1.2 Hardware wiring diagrams	3 days	Mon 5/4/26	Wed 5/6/26
83		6.1.3 Firmware documentation	3 days	Thu 5/7/26	Sat 5/9/26
84		6.2 User Documentation	6 days	Sun 5/10/26	Fri 5/15/26
85		6.2.1 Setup instructions	2 days	Sun 5/10/26	Mon 5/11/26
86		6.2.2 Calibration guidelines	2 days	Tue 5/12/26	Wed 5/13/26
87		6.2.3 Dashboard usage guide	2 days	Thu 5/14/26	Fri 5/15/26
88		7. Deployment & Handover	3 days	Sat 5/16/26	Tue 5/19/26
89		7.1 Final Prototype	2 days	Sat 5/16/26	Mon 5/18/26
90		7.1.1 Install system on-site	1 day	Sat 5/16/26	Sat 5/16/26
91		7.1.2 Configure dashboard	1 day	Sun 5/17/26	Sun 5/17/26
92		7.1.3 Test final functionality	1 day	Mon 5/18/26	Mon 5/18/26
93		7.2 Project Handover	1 day	Tue 5/19/26	Tue 5/19/26
94		7.2.1 Final report submission	1 day	Tue 5/19/26	Tue 5/19/26
95		7.2.2 Presentation and demonstration	1 day	Tue 5/19/26	Tue 5/19/26
96		7.2.3 Future recommendations	1 day	Tue 5/19/26	Tue 5/19/26
97		Milestone 5: Project Closeout	1 day	Tue 5/19/26	Tue 5/19/26

Figure 5 Gantt Chart (part-2)

Table 2 Gantt Chart Timeline

Name	Duration	Start Date	End Date
1. Project Initiation & Planning	**26 days**	**9-Nov-25**	**4-Dec-25**
1.1 Requirement Gathering	**12 days**	**9-Nov-25**	**24-Nov-25**
1.1.1 Identify user needs	4 days	9-Nov-25	12-Nov-25
1.1.2 Define system functionalities	4 days	13-Nov-25	16-Nov-25
1.1.3 Gather agricultural sensor requirements	4 days	17-Nov-25	20-Nov-25
1.1.4 Conduct feasibility analysis	4 days	21-Nov-25	24-Nov-25
1.2 Project Planning	**10 days**	**25-Nov-25**	**4-Dec-25**
1.2.1 Create project scope	3 days	25-Nov-25	27-Nov-25
1.2.2 Define project deliverables	2 days	28-Nov-25	29-Nov-25
1.2.3 Develop incremental plan	2 days	30-Nov-25	1-Dec-25
1.2.4 Define risks and mitigation	3 days	2-Dec-25	4-Dec-25
Milestone 1: Planning Complete	**0 days**	**4-Dec-25**	**4-Dec-25**
2. System Design	**35 days**	**5-Dec-25**	**8-Jan-26**
2.1 System Architecture	**20 days**	**5-Dec-25**	**24-Dec-25**
2.1.1 Hardware architecture diagram	5 days	5-Dec-25	9-Dec-25
2.1.2 Software architecture diagram	5 days	10-Dec-25	14-Dec-25
2.1.3 Data flow design	4 days	15-Dec-25	18-Dec-25
2.1.4 Communication model design	6 days	19-Dec-25	24-Dec-25
2.2 UI/UX Design	**15 days**	**25-Dec-25**	**8-Jan-26**
2.2.1 Dashboard wireframes	5 days	25-Dec-25	29-Dec-25
2.2.2 Display layout planning	5 days	30-Dec-25	3-Jan-26
2.2.3 Alert interface design	5 days	4-Jan-26	8-Jan-26
Milestone 2: Architecture Approved	**0 days**	**8-Jan-26**	**8-Jan-26**

3. Hardware Development	**51 days**	**9-Jan-26**	**28-Feb-26**
3.1 Microcontroller Setup	**12 days**	**9-Jan-26**	**20-Jan-26**
3.1.1 ESP32 configuration	4 days	9-Jan-26	12-Jan-26
3.1.2 Power supply setup	4 days	13-Jan-26	16-Jan-26
3.1.3 Battery and buck converter	4 days	17-Jan-26	20-Jan-26
3.2 Sensor Integration	**25 days**	**21-Jan-26**	**14-Feb-26**
3.2.1 Soil moisture sensor setup	4 days	21-Jan-26	24-Jan-26
3.2.2 DHT22 sensor configuration	4 days	25-Jan-26	28-Jan-26
3.2.3 BH1750 light sensor integration	4 days	29-Jan-26	1-Feb-26
3.2.4 pH sensor connection	5 days	2-Feb-26	6-Feb-26
3.2.5 NPK sensor and RS485 setup	8 days	7-Feb-26	14-Feb-26
3.3 Hardware Assembly	**14 days**	**15-Feb-26**	**28-Feb-26**
3.3.1 Mount sensors on frame	4 days	15-Feb-26	18-Feb-26
3.3.2 Prepare wiring and connectors	4 days	19-Feb-26	22-Feb-26
3.3.3 Install enclosure and protection	4 days	23-Feb-26	26-Feb-26
3.3.4 Field deployment setup	2 days	27-Feb-26	28-Feb-26
Milestone 3: Hardware Ready	**0 days**	**28-Feb-26**	**28-Feb-26**
4. Incremental Software	**67 days**	**1-Mar-26**	**15-May-26**
4.1 Increment 1 – Basic Functionality	**19 days**	**1-Mar-26**	**19-Mar-26**
4.1.1 Firmware for moisture	5 days	1-Mar-26	5-Mar-26
4.1.2 Firmware for Temp/Humidity	5 days	6-Mar-26	10-Mar-26
4.1.3 Firmware for Sunlight	5 days	11-Mar-26	15-Mar-26
4.1.4 Basic data print on Serial	4 days	16-Mar-26	19-Mar-26
4.2 Increment 2 – Data Processing	**24 days**	**20-Mar-26**	**12-Apr-26**
4.2.1 pH sensor calibration	5 days	20-Mar-26	24-Mar-26
4.2.2 NPK sensor calibration	6 days	25-Mar-26	30-Mar-26

4.2.3 Noise reduction algorithms	6 days	31-Mar-26	5-Apr-26
4.2.4 Multi-sensor data integration	7 days	6-Apr-26	12-Apr-26
4.3 Increment 3 – Data Visualization	**20 days**	**13-Apr-26**	**2-May-26**
4.3.1 Real-time data interface	6 days	13-Apr-26	18-Apr-26
4.3.2 Graphs and visual indicators	6 days	19-Apr-26	24-Apr-26
4.3.3 Historical data logging	5 days	25-Apr-26	29-Apr-26
4.3.4 MicroSD/Cloud sync setup	3 days	30-Apr-26	2-May-26
4.4 Increment 4 – Alerts & Logic	**13 days**	**3-May-26**	**15-May-26**
4.4.1 Threshold-based alert logic	4 days	3-May-26	6-May-26
4.4.2 Soil condition recommendations	5 days	7-May-26	11-May-26
4.4.3 Notification integration	4 days	12-May-26	15-May-26
4.5 Increment 5 – Final Integration	**16 days**	**16-Apr-26**	**1-May-26**
4.5.1 Integrate all software modules	7 days	16-Apr-26	22-Apr-26
4.5.2 Optimize performance	4 days	23-Apr-26	26-Apr-26
4.5.3 Debug and fix critical issues	5 days	27-Apr-26	1-May-26
Milestone 4: Software Complete	**0 days**	**15-May-26**	**15-May-26**
5. Testing & Validation	**17 days**	**2-May-26**	**18-May-26**
5.1 Unit Testing	**8 days**	**2-May-26**	**9-May-26**
5.1.1 Sensor-level testing	3 days	2-May-26	4-May-26
5.1.2 Microcontroller firmware testing	3 days	5-May-26	7-May-26
5.1.3 UI testing	2 days	8-May-26	9-May-26
5.2 Integration Testing	**3 days**	**10-May-26**	**12-May-26**
5.2.1 Hardware and software integration	1 day	10-May-26	10-May-26
5.2.2 Data consistency testing	1 day	11-May-26	11-May-26
5.2.3 Wireless communication testing	1 day	12-May-26	12-May-26
5.3 Field Testing	**6 days**	**13-May-26**	**18-May-26**

5.3.1 Soil data accuracy testing	2 days	13-May-26	14-May-26
5.3.2 Environmental stress testing	2 days	15-May-26	16-May-26
5.3.3 User evaluation and feedback	2 days	17-May-26	18-May-26
6. Documentation	**15 days**	**1-May-26**	**15-May-26**
6.1 Technical Documentation	**9 days**	**1-May-26**	**9-May-26**
6.1.1 System architecture documentation	3 days	1-May-26	3-May-26
6.1.2 Hardware wiring diagrams	3 days	4-May-26	6-May-26
6.1.3 Firmware documentation	3 days	7-May-26	9-May-26
6.2 User Documentation	**6 days**	**10-May-26**	**15-May-26**
6.2.1 Setup instructions	2 days	10-May-26	11-May-26
6.2.2 Calibration guidelines	2 days	12-May-26	13-May-26
6.2.3 Dashboard usage guide	2 days	14-May-26	15-May-26
7. Deployment & Handover	**4 days**	**16-May-26**	**19-May-26**
7.1 Final Prototype	**3 days**	**16-May-26**	**18-May-26**
7.1.1 Install system on-site	1 day	16-May-26	16-May-26
7.1.2 Configure dashboard	1 day	17-May-26	17-May-26
7.1.3 Test final functionality	1 day	18-May-26	18-May-26
7.2 Project Handover	**1 day**	**19-May-26**	**19-May-26**
7.2.1 Final report submission	1 day	19-May-26	19-May-26
7.2.2 Presentation and demonstration	1 day	19-May-26	19-May-26
7.2.3 Future recommendations	1 day	19-May-26	19-May-26
Milestone 5: Project Closeout	**0 days**	**19-May-26**	**19-May-26**

10.Conclusion

The Smart Agriculture Monitoring & Management System (AgroBot) represents comprehensive integration of IoT and AI where the IoT sensors collect the data and AI analyses the data. The continuous real time data, insights and recommendations help farmers optimize resource while maximizing crop yield.

With global agriculture sector moving toward the new concept of precise agriculture farming and related techniques, this project presents itself as practical, cost-effective in the long run and a smart solution for the farmers. Upon successful implementation of the system and further development of its future adaptable features this system will serve as a foundation for smart farming initiative impacting livelihood for the farming community in our society all over the world.